College Board AP® Chemistry

Your notes

Chemical Bonding

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- Valence Electrons in a Metallic Solid
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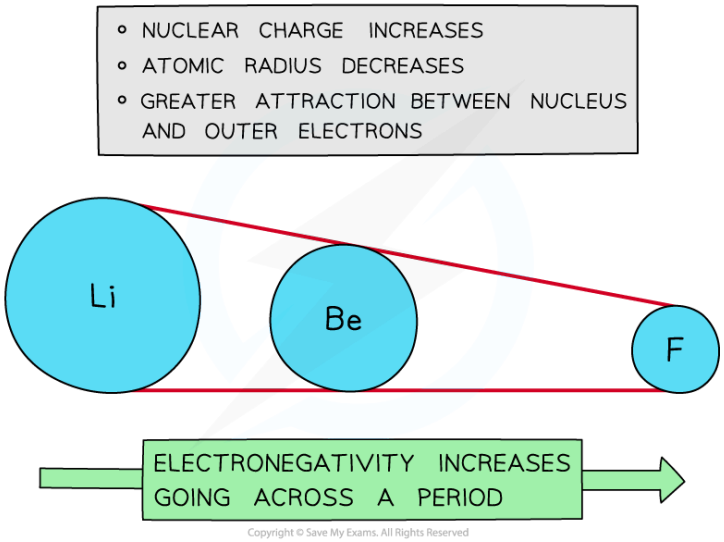
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Electronegativity values

- **Electronegativity** is the tendency of an atom to attract a pair of electrons when it forms a covalent bond, mainly involving valence electrons
- Electronegativity generally increases across a period
  - Since the number of protons increase, there **effective nuclear charge increase**
  - This makes the **Coulombic attraction** between the nucleus and the electrons from the covalent bond **stronger**
  - Therefore, the **electronegativity value will increase**

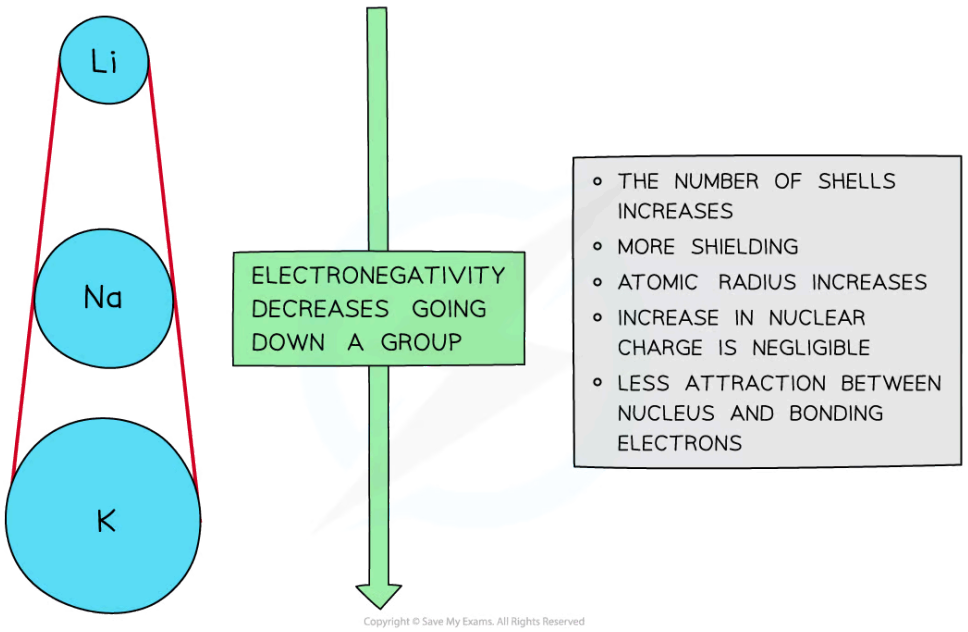
Change in electronegativity across a period



Electronegativity increases going across the periods of the periodic table

- Down a group, electronegativity decreases
  - The main responsible is the **increase in the shielding effect**
  - Since the **distance** between the nucleus and the valence electrons increases, the **Coulombic forces attraction are considerably weaker**
  - In consequence, the **effective nuclear charge decreases,** the electrons from a covalent bond are weakly attracted, and the **electronegativity value decrease**

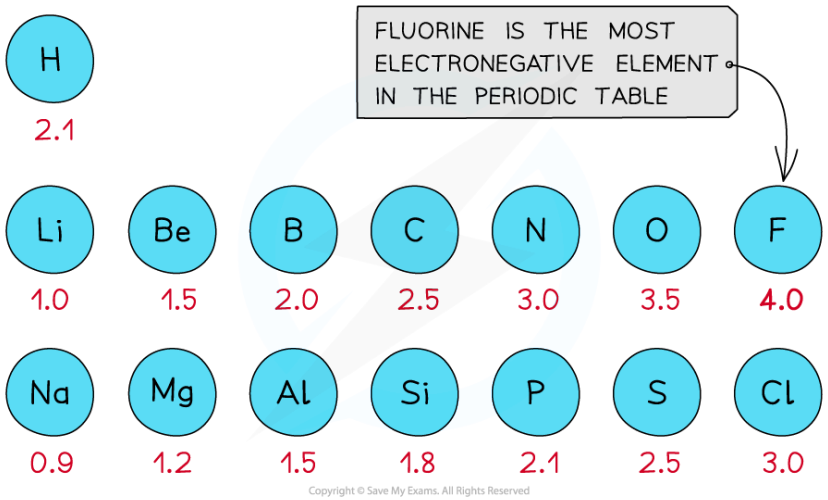
Change in electronegativity down a group



Electronegativity decreases going down the groups of the Periodic table

- The **Pauling scale** has the values of electronegativity for each atom
  - Noble gasses does not appear in the Pauling scale since they do not form covalent bonds because of their **full valence shell**

The Pauling Scale



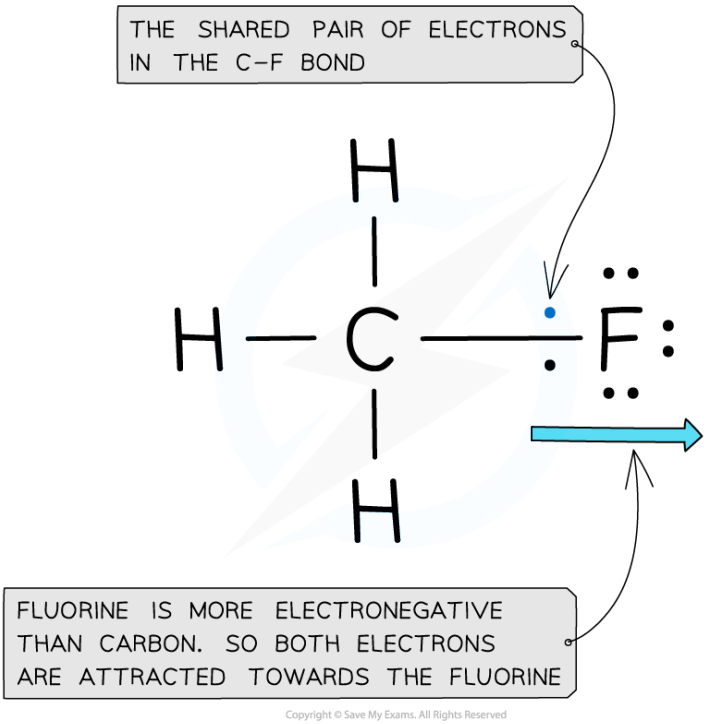
First three rows of the periodic table showing electronegativity values

- Fluorine (F) is the most electronegative atom in the periodic table
  - It is the best when attracting electron density towards itself when it forms a covalent bond with an atom

Your notes

Molecule of CH_3F

Your notes



Electron distribution in the C-F bond of fluoromethane

Your notes

Non-polar Covalent bonds

- A **covalent bond** is formed when two **nonmetals** share a pair of electrons
- Atoms form covalent bonds to gain a **full outer shell** of electrons
- Compounds that are formed by elements joined by covalent bonds are **molecules**
- Covalent bonds come in different presentations:
  - A **single covalent bond** occurs when **one pair** of electrons is shared, and each atom contributes with one valence electron
  - A **double covalent bond** occurs when **two pairs** of electrons are shared, and each atom contributes with two valence electrons
  - A **triple covalent bond** occurs when **three pairs** of electrons are shared, and each atom contributes with three valence electrons
- The nature of the covalent bond depends in the **difference of electronegativity** between the atoms that share the pairs of electrons
  - Using the Pauling's scale of electronegativity, **if the difference is smaller than 0.4, the covalent bond is nonpolar**
- **Nonpolar covalent bonds** are formed when two elements with **similar or equal electronegativity** share pairs of electrons
  - E.g. Chlorine gas is a **diatomic element**. Therefore, it exists in nature as a molecule of Cl_2 instead of an individual atom Cl
  - The two chlorine atoms form a single covalent bond sharing a pair of electrons as shown in the image below
  - Since both atoms are the same, their difference in electronegativity is 0
  - Therefore, the nature of the covalent bond is nonpolar

The Formation of a Chlorine Molecule, Cl_2

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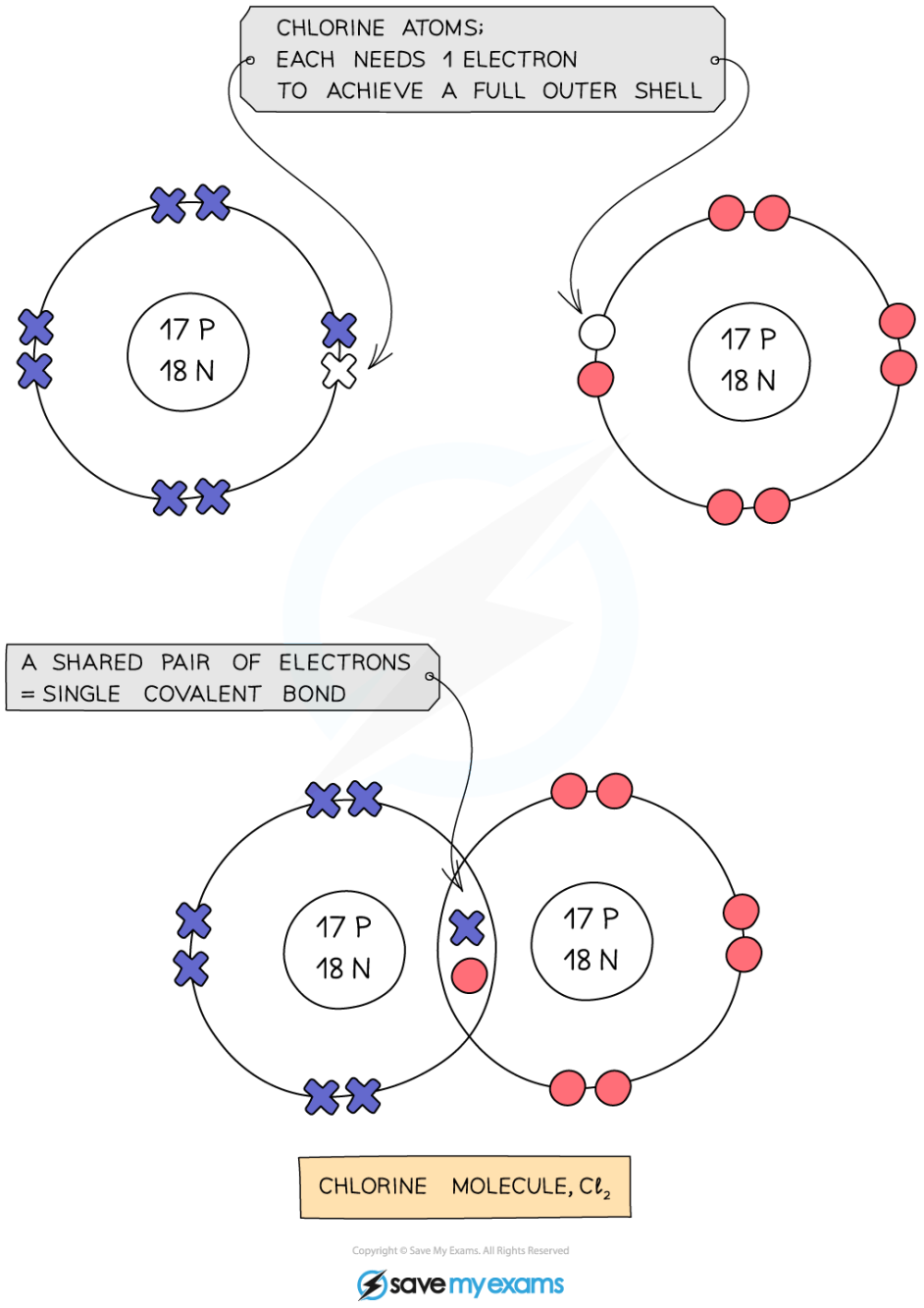


Diagram showing how a covalent bond forms between two chlorine atoms

Examiner Tips and Tricks

The known diatomic elements are: H_2, N_2,F_2, O_2, I_2, Cl_2, Br_2. You can remember them by using the mnemonic: **H**ave **N**o **F**ear **O**f **I**ce **C**old **B**ear

Polar covalent bonds

Your notes

- A **polar covalent bond** is formed when the nonmetals that share pairs of electrons have a greater electronegativity difference
  - Using the Pauling's scale of electronegativity, **if the difference is between than 0.4 - 2.0, the covalent bond is polar**
- **Electronegativity** is ability of an atom to attract the pair electrons of a covalent bond
- One of the atoms will have greater Coulombic attraction to the electrons, pulling them towards itself
- When this occurs, a **dipole moment (δ)** is formed because there was a separation of positive and negative charges
- A dipole moment is created when particles of equal magnitude but opposite sign charges are separated
  - E.g. This occurs in the molecule of hydrofluoric acid (HF) that is shown in the image below
  - The chlorine atom and the hydrogen atom form a single covalent bond sharing a pair of electrons as shown in the image below
  - Their difference in electronegativity is calculated as following

$$\Delta \text{ electronegativity} = \text{Highest electronegativity} - \text{Lowest electronegativity}$$

$$\Delta \text{ electronegativity} = 4.0 - 2.1$$

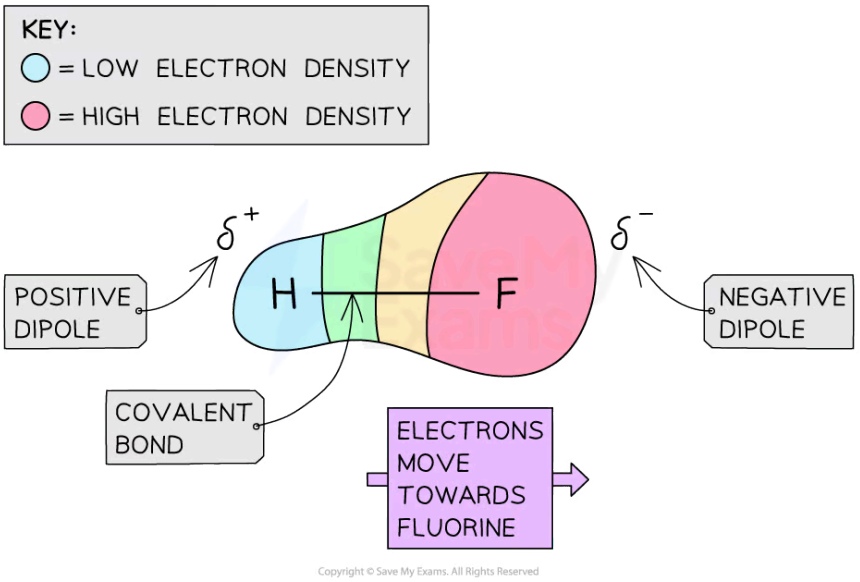
$$\Delta \text{ electronegativity} = 1.9$$

- Since the difference is between 0.4 and 2.0, the nature of the covalent bond is **polar**
- Fluorine will acquire a negative dipole **δ^-**
- Hydrogen will acquire a negative dipole **δ^+**
- If the difference in electronegativity is **greater**, the bond dipoles are **greater too**
  - This occurs because one of the atoms is pulling the electrons with stronger coulombic attrraction, therefore the distance between the charges is closer

The H-F molecule

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Polar covalent bond in hydrofluoric acid (HF)

Ionic character

- **Ionic bonds** are formed when electrons are transferred from one atom to another
- In polar covalent bonds, there is a partial transfer of electrons towards the most electronegative atom
- Therefore, **polar covalent bonds** have some ionic character
- The **ionic character** of a covalent bond increase if the difference of electronegativity between the atoms is greater

Your notes

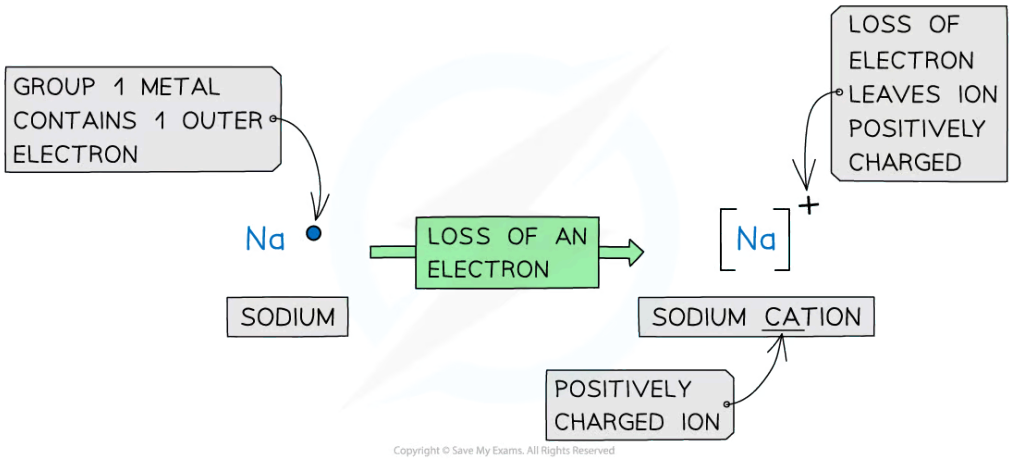
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Ionic vs Covalent

Ionic bonding

- An **ionic bond** is formed when a metal **donates electrons** to a non metal
  - The **metal** becomes a **positive ion** (cation) because it has lost electrons

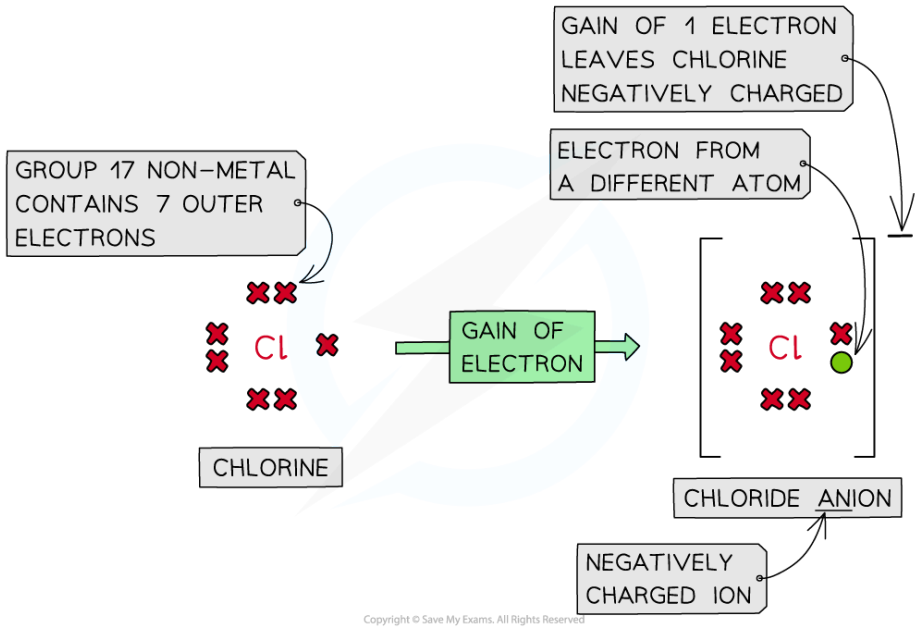
Forming a Cation



Forming cations by the removal of electrons from metals

- The **non metal** becomes a **negative ion** (anion) because it has gained electrons

Forming an Anion

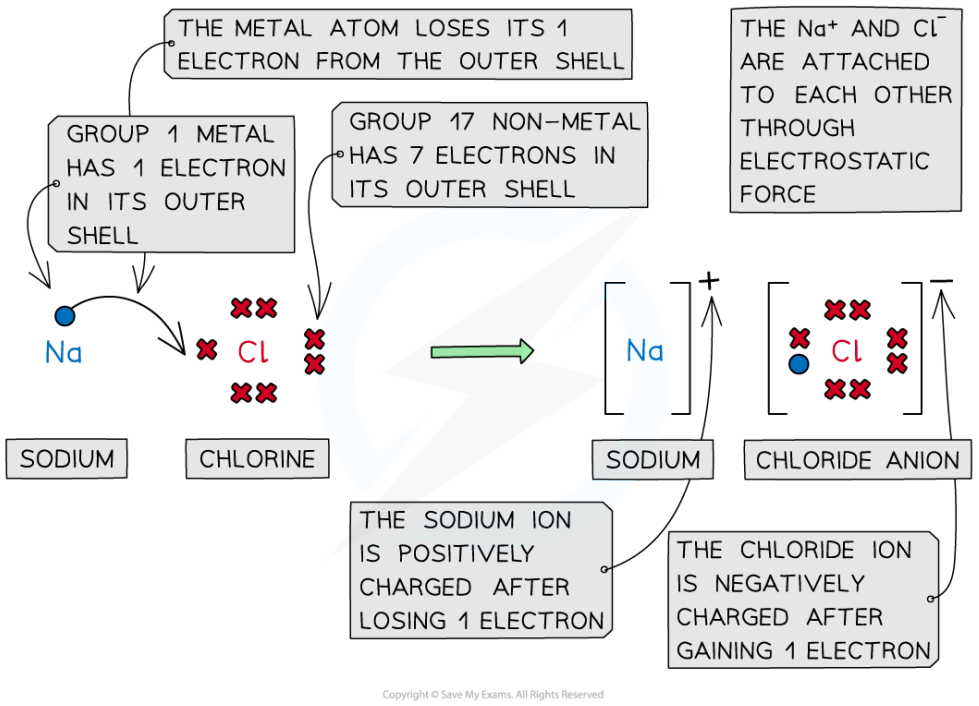


Forming anions by the addition of electrons to nonmetals

Your notes

- The ionic bond is formed because metals and the non metals want to have a full outer shell of electrons
- The **difference in electronegativity**, using the Pauling's scale, between the atoms that are part of an ionic bond is **greater than 2.0**
- A **full outer shell implies more stability,** that is why noble gasses do not form ions
- When the ionic bond is formed, their electron configurations are the same as their closest noble gas. Sodium chloride will be used as the main example below
  - Na has **one valence electron**. Na in sodium chloride (NaCl) loses one electron to form Na^+ Therefore, Na^+ has the electron configuration $1s^2\ 2s^2\ 2p^6$, which is the same as Ne
  - Cl has **seven valence electrons**. Cl in sodium chloride (NaCl) gains one electron to form Cl^- Therefore, Cl^- has the electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$, which is the same as Ar
  - When ions are formed, **an strong Coulombic force of attraction** is formed between the oppositely charged ions

Forming an Ionic Bond



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Cations and anions bond together using strong Coulombic forces of attraction, which require a lot of energy to overcome

Covalent bonding

- A **covalent bond** is formed when two **nonmetals** share a pair of electrons

- Atoms form covalent bonds to gain a **full outer shell** of electrons
- There are two types of covalent bonds: **polar** and **nonpolar**
  - This classification depends on the difference in electronegativities between the atoms in the bond



Your notes

Ionic vs covalent compounds

- The best way to differentiate between ionic or covalent compounds is analyzing the **physical** and **chemical** properties

Table comparing the properties between ionic and covalent compounds

	Ionic compounds	Covalent compounds
Melting point/Boiling point	High	Low
Electrical conductivity	Only when molten or in solution	Do not conduct electricity
Solubility	Generally soluble in water	Usually insoluble unless they are **polar covalent**
Hardness	Hard, brittle	Soft
Physical state at room temperature	Solid	Solid, liquid or gasses depending on their **intermolecular forces**
Forces	Strong coulombic force of attraction between cations and anions	Weak forces between the molecules
Difference in electronegativity	Greater than 2.0	**Polar covalent:** Between 0.4 and 2.0 **Non polar covalent:** Lower than 0.4
Examples	NaCl	**Polar covalent:** HF **Non polar covalent:** H_2



Examiner Tips and Tricks

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Make sure to know how to differentiate the properties of ionic and covalent compounds since this section is usually assessed in Section 1 of your AP Chemistry Exam



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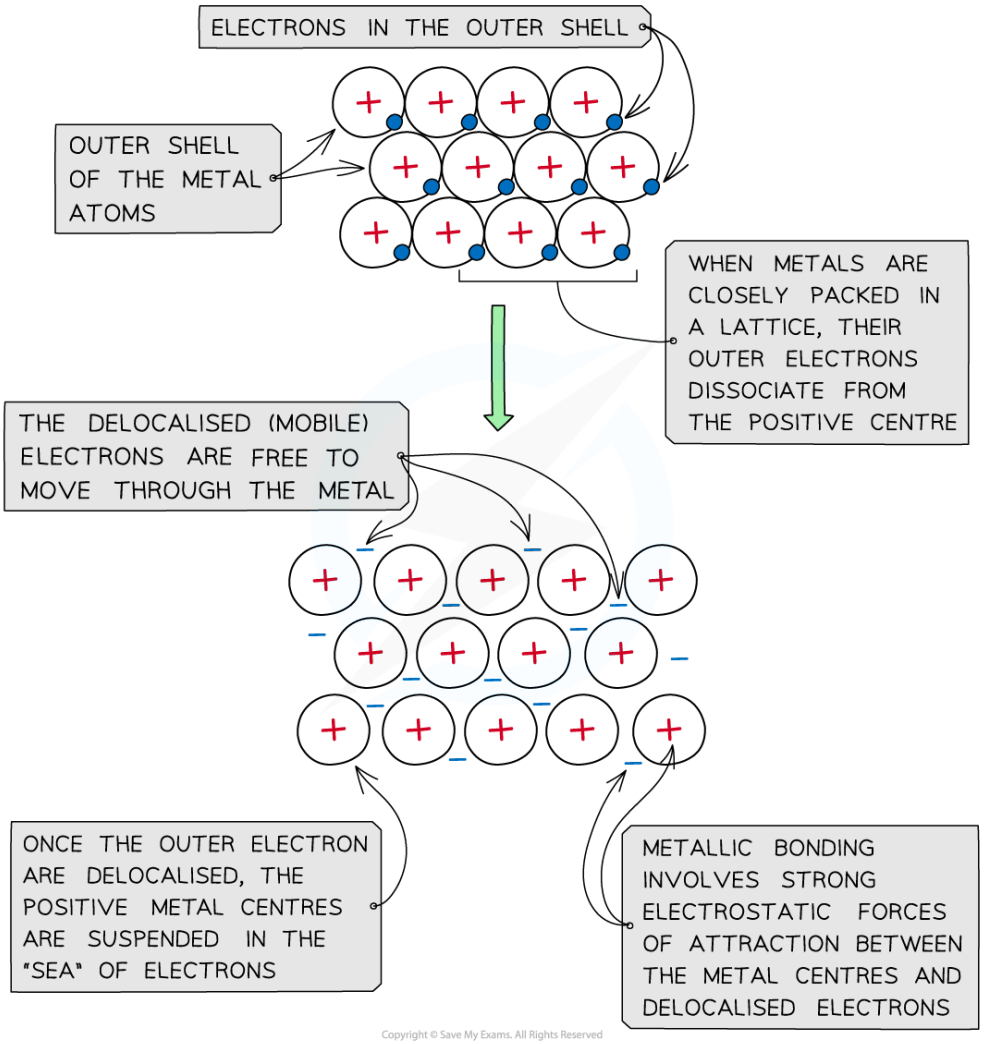
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Valence electrons in a Metallic Solid

- The atoms in a metallic solid are arranged together in **metallic lattices**
- Since the atoms are tightly packed together, they lose valence electrons and become positive ions
- Therefore, **electrons are free to move** throughout the structure
  - The free-moving electron are called **"delocalized"** because they do not correspond to particular atoms anymore
- The **metallic solids** are held together by the Coulombic forces of attraction between the **positively charged metal ions** and a **sea of delocalized electrons**
- The interaction between ions and electrons is named **metallic bonding**

The Structure of a Metal



The positive metal centers interact with a Coulombic force of attraction with in a 'sea' of delocalised electrons



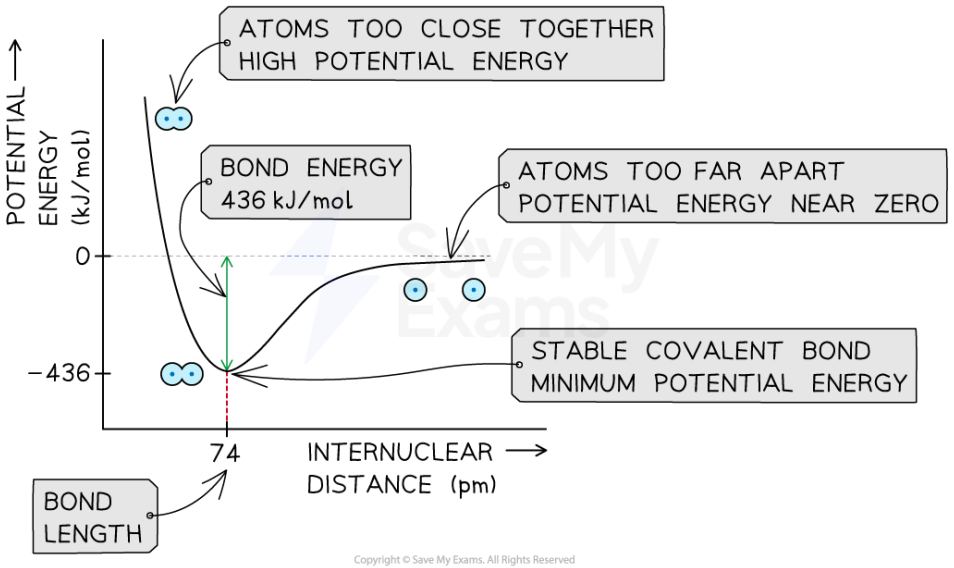
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Potential Energy & Interatomic Distance

- A potential energy diagram represents the potential energy as a function of internuclear distance
  - The distance at which the potential energy is at a minimum is the **bond length** of the covalent bond
  - The difference between the minimum potential energy and zero is the **bond energy**, the amount of energy required to break the covalent bond

Potential Energy Diagram



Potential energy diagram of hydrogen gas (H_2) showing a minimum potential energy of -436 kJ/mol which corresponds to a bond length of 74 pm

- When two atoms are far away
  - There is little attraction between the positively charged nucleus of one atom and the negatively charged electrons of the other atom
  - The potential energy is **near zero**
- When two atoms are very close
  - The two positively charged nuclei and the two negatively charged electron clouds strongly repel one another
  - The potential energy is **very high**
- When two atoms form a stable covalent bond
  - The attractive and repulsive forces between the two atoms are balanced

- The potential energy is at a **minimum**.

Your notes

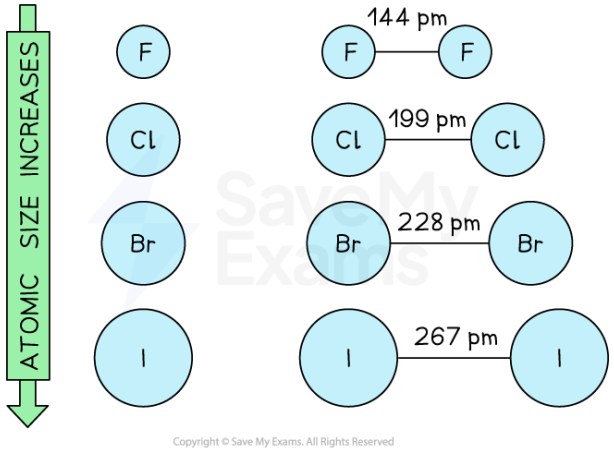
Examiner Tips and Tricks

The minimum potential energy will always be a negative value. Bond energy is always a positive value as breaking chemical bonds is an endothermic process.

Factors Influencing Bond Length

- As atomic radius increases, bond length increases
  - Bond length is typically reported in units of picometers (pm)
  - A picometer is 1×10^{-12}m

Bond length in halogens



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Diagram showing that the bond length of diatomic halogen molecules increases with increasing atomic radius

- **Bond order** indicates the number of electron pairs that are shared between two atoms
- As bond order increases, bond length decreases while bond energy increases

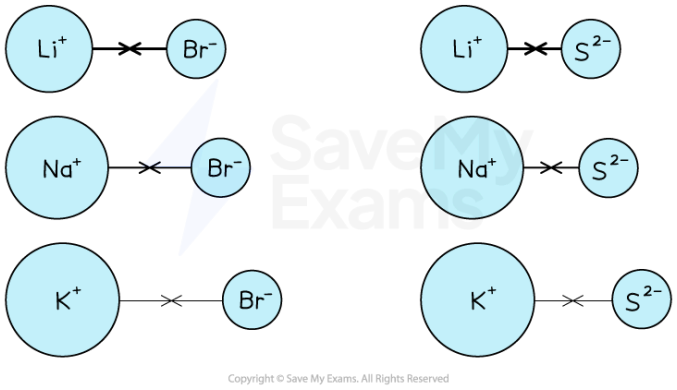
Bond	Bond Order	Number of shared electron pairs	Bond length (pm)	Bond energy (kJ/mol)
C-C	1	1	154	347
C=C	2	2	133	620
C≡C	3	3	120	812

Your notes

Coulomb's Law & Attractive Forces

- Coulomb's law describes the electrostatic force between two charged particles
- The formula for Coulomb's law is $F_{coulombic} = k\frac{q_1 q_2}{r^2}$, where:
  - $F_{coulombic}$ is the force between the charges
  - k is Coulomb's constant
  - q_1 and q_2 are the value of the charges
  - r is the distance between the charges
- For a cation and anion in an ionic bond, $F_{coulombic}$
  - **increases** with **increasing charge** of each ion
  - **decreases** with **increasing ionic radius**

Coulombic force between cations and anions

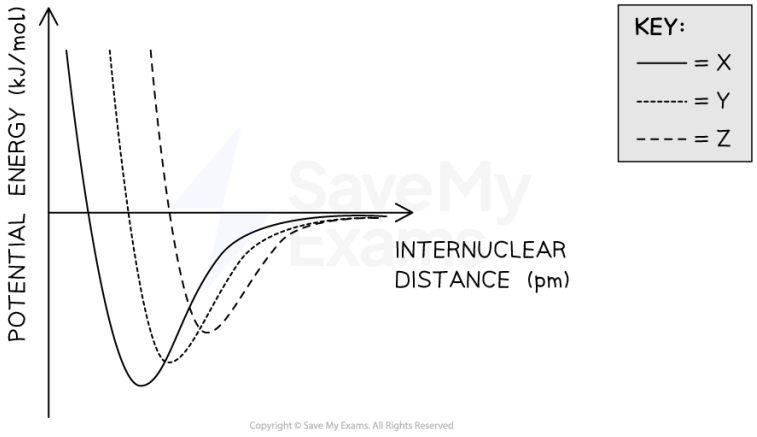


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Diagram showing that the Coulombic force between a cation and anion decreases with increasing ionic radius and increases with increasing charge

- The **greater the force** between the charges
  - the stronger the interaction between the ions
  - the greater the bond energy (also known as **lattice energy**)
  - the **more negative the minimum potential energy** of a stable ionic bond between the two ions

Worked Example



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The graph shows the potential energy of three potassium salts, X, Y, and Z, as a function of internuclear distance. Based on the data, what are the most likely identities of compounds X, Y, and Z?

	X	Y	Z
A.	KI	KBr	KCl
B.	KBr	KCl	KI
C.	KCl	KI	KBr
D.	KCl	KBr	KI

Answer:

- Each potassium salt contains a K^+ ion and a halide ion with a 1– charge
  - So, we do not need to consider the effect that charge has on internuclear distance or potential energy
- The ionic radius of the halide ions increases down the group, thus $Cl^- < Br^- < I^-$
- As ionic radius increases
  - the internuclear distance between the potassium and halide ions increases
  - $F_{coulombic}$ between the ions decreases
  - the lattice energy decreases
  - the potential energy minimum becomes less negative
- So, KCl with the smallest halide ion will have the smallest internuclear distance and the most negative minimum potential energy
- KI with the largest halide ion will have the largest internuclear distance and the least negative minimum potential energy
- Therefore, the answer is **D**